

7/3/23

**SASTRA DEEMED UNIVERSITY**

(A University under section 3 of the UGC Act, 1956)

**End Semester Examinations**

**JULY 2023**

**Course Code: PHY101R01**

**Course: ENGINEERING PHYSICS**

**QP No. :UF021-2**

**Duration: 3 hours**

**Max. Marks:100**

**PART-A**

**Answer all the questions**

**10 x 2 = 20 Marks**

1. Find the K.E. (in eV) of a neutron with a de Broglie wavelength of 0.3 nm. Is the relativistic calculation needed?
2. Why wave function of the particle present outside the potential box is zero?
3. Compare the uncertainties in velocity of a proton and an electron contained in a  $10 \text{ \AA}$  box.
4. List any two applications of LASER in the field of medicine.
5. Choose any two pumping methodologies available and write short note on it with suitable examples.
6. Calculate the refractive indices of the cladding and core having  $NA = 0.21$  and  $\Delta = 0.011$ .
7. Compare polar and non-polar dielectrics.
8. Write down any two applications of magnetic materials.

9. Determine the number density of donor atoms which have to be added to an intrinsic semiconductor to produce an n – type semiconductor of conductivity  $5 \Omega^{-1}\text{cm}^{-1}$ . Given the mobility of electrons is  $3850 \text{ m}^2.\text{V}^{-1}.\text{s}^{-1}$ .
10. How can you differentiate conductors, n – type semiconductors and p – type semiconductors using hall coefficients.

### PART- B

**Answer any FOUR the questions**

**4 x 15 = 60 Marks**

11. Derive an equation, which determines the behavior of particles on a subatomic level.
12. Draw the diagram of gaseous laser, explain its construction, working and features of it.
13. Compare different types of fibers based on the mode of probation of signal through it.
14. Derive an equation which relates  $\epsilon_r$  and  $\alpha_c$  of a dielectric material.
15. Compare different types magnetic materials you have studied.
16. Derive an expression to find the carrier concentration of intrinsic semiconductors.

### PART-C

**Answer the following**

**1 x 20 = 20 Marks**

17. An electron is bound by a potential box of infinite height having width of  $4.6 \text{ \AA}$ . Find the energy of a particle when its Eigen functions have 1 and 2 antinodes and also find the carrier concentration of charge carriers at room temperature, if the RH value is  $3.6 \times 10^{-4} \text{ m}^3/\text{C}$ . If the box is confined to a new dimension of  $3.5 \text{ \AA}$ , find whether if there is any change in carrier concentration and energy gap between

the energy levels when its Eigen functions have 1 and 2 antinodes. Also find the shift in wavelength takes place due to change in dimension, if the electron comes down to least energy level from 2nd antinode. The frequencies found above is made to pass through a dielectric material, whose refractive index is 1.5. Calculate the electronic polarization of the material and find whether there will be a dielectric loss when the above found frequencies were passing through it.

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